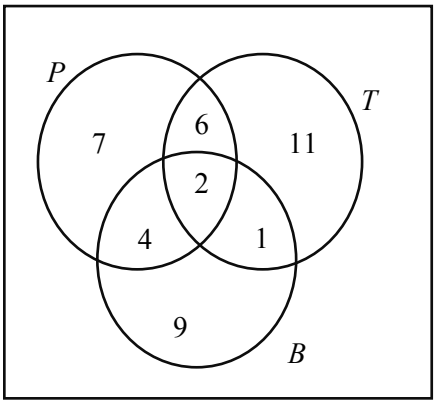
Question	Answer	Marks	Partial Marks
9(a)		**3**	**B2** for 5 correct entries including '2' correctly placed at the intersection of the 3 sets or **M1** for $k+8-k+3-k+6-k=40-(7+9+11)$ oe or for k, $8-k$, $3-k$, $6-k$, seen correctly placed on diagram with 7, 11 and 9 correctly placed
9(b)	11	**1**	
9(c)	Ø or { }	**1**	
9(d)	$\frac{7}{260}$ oe	**2**	**M1** for $\frac{7}{40}\times\frac{6}{39}$ oe
9(e)	$\frac{14}{95}$ oe	**2**	**FT** *their* Venn diagram **M1** for $\frac{8}{20}\times\frac{7}{19}$
10(a)(i)	$4x-13$ final answer	**1**	
10(a)(ii)	$25x^2$ final answer	**1**	
10(b)	$\frac{x+1}{4}$ or $\frac{x}{4}+\frac{1}{4}$	**2**	**M1** for correct first step $x=4y-1$ or $y+1=4x$ or $\frac{y}{4}=x-\frac{1}{4}$
10(c)	0.6934 final answer	**3**	**B2** for 0.69336… or $3^{-\frac{1}{3}}$ oe or 0.693 or **M1** for $3^{-3^{-x}}$ oe
10(d)(i)	$(3x-2)^2-3^{-(-3)}$	**M1**	
	$9x^2-6x-6x+4-27$ or $9x^2-12x+4-27$ leading to $9x^2-12x-23$	**A1**	with no errors seen

Question	Answer	Marks	Partial Marks
10(d)(ii)	$\dfrac{-(-12)\pm\sqrt{(-12)^2-4(9)(-23)}}{2\times 9}$ or better	**B2**	**B1** for $\sqrt{(-12)^2-4(9)(-23)}$ oe or $\dfrac{-(-12)+\sqrt{q}}{2\times 9}$ oe or $\dfrac{-(-12)-\sqrt{q}}{2\times 9}$ oe or both
	– 1.07, 2.40 final answers	**B2**	**B1** for each If **B0, SC1** for answers – 1.1 or –1.06 or –1.065… to – 1.065 **and** 2.4 or 2.39 or 2.398 to 2.398… or – 1.07 **and** 2.40 seen in working or for –2.40 and 1.07 as final answer
10(e)	-5 final answer	**2**	**M1** for $243 = 3^{-x}$
11(a)	(1, 2) (–1, 6)	**5**	**B2** for [derivative oe =] $3x^2-3$ or **B1** for [derivative oe =] $3x^2$ or $f(x)-3$ **M1** for *their* derivative = 0 or recognition of $\dfrac{dy}{dx} = 0$ oe **B1** for [x =] –1, 1 or for one coordinate pair
11(b)	(1, 2) minimum with reason (–1, 6) maximum with reason	**3**	Reasons could be e.g. a reasonable sketch correct use of 2nd derivative = $6x$ = 6 , 6 > 0, so (1, 2) minimum oe 2nd derivative = $6x$ = –6 , –6 < 0 so (–1, 6) maximum oe, or finds gradient on each side of both correct stationary points with correct conclusion **B2** for 1 correct with reason or **M1** for showing [2nd derivative =] $6x$ or gradients for one value on either side of one correct stationary point or for reasonable sketch of cubic